

MUHAMMAD AHMED MUFTI

Backend Engineer

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SUMMARY

Backend engineer working in Node.js, Python, and Java. Shipped the APIs behind three production platforms (PostgreSQL, DuckDB, MongoDB), built an event-driven order pipeline on Spring Boot and Kafka with Testcontainers integration tests, and pen-tested enterprise web applications as a security intern. Computer Science graduate, FAST-NUCES (CGPA 3.6 / 4.00).

TECHNICAL SKILLS

Languages	JavaScript / TypeScript (Node.js), Python, Java, SQL, C/C++
Frameworks	Express, FastAPI, Spring Boot, Spring Kafka, Mongoose, Socket.io, LangChain
Data & Messaging	PostgreSQL, MySQL, SQL Server (T-SQL), MongoDB, DuckDB, SQLite, Apache Kafka, FAISS
API & Security	REST design, OpenAPI, JWT, RBAC, bcrypt, validation, rate limiting, OWASP Top 10, AES / RSA / X.509
Testing & Ops	JUnit 5, Testcontainers, Postman, Docker Compose, Git, Linux, Modal (serverless GPU), Vercel

SELECTED PROJECTS

Order Pipeline: Two Spring Boot Services over Kafka 2026

Java, Spring Boot, Spring Kafka, Kafka 3.9 (KRaft), Docker Compose, Testcontainers.

- order-service accepts POST /orders and publishes OrderCreated events to a three-partition topic; inventory-service consumes them, reserves stock, deduplicates by event id, and routes malformed or non-retryable events to a dead-letter topic.
- Verified consumer-group rebalancing across two instances, replay after a consumer restart with no reprocessing of committed offsets, fan-out to a second consumer group, and poison-pill handling; integration tests run against Kafka in Testcontainers.

recs.co: Sentiment Pipeline and API LIVE 2025

Python, FastAPI, DuckDB, Modal GPU (A100 / T4), fine-tuned RoBERTa. Live: recs.co

- Classified 14.1 million Reddit comments from 937 subreddits with a fine-tuned RoBERTa model on Modal GPU workers and served product rankings, grades, and guides through a FastAPI and DuckDB API.

MedScholar: Retrieval-Augmented Clinical Research Service LIVE 2025 - 2026

FastAPI, PostgreSQL, LangChain, FAISS, PubMed API. Live: med-scholar-ten.vercel.app

- Backend that pulls literature from PubMed, indexes it in FAISS, and answers clinical questions with cited sources under strict grounding constraints to limit hallucination; final year project.

Pakistan Legal Nexus: Legal Operations API LIVE 2025 - 2026

Node.js, Express, MongoDB / Mongoose, JWT, Socket.io, Multer. Live: earnest-praline-2c72a6.netlify.app

- REST and WebSocket back end with 83 endpoints over 16 models: role-based access for lawyers, litigants, and clerks, case files with document upload, real-time chat rooms, and a Hugging Face API integration.

Inventory Control Database 2026

SQL Server, T-SQL. Code: github.com/AhmedMufti/SQL-Server-inventory-control-database

- Trigger-driven stock ledger that updates quantities on every movement and blocks negative stock, with stored procedures for stock in and out, monthly movement reports, and low-stock alerts; seeded with 20 products and two months of transactions.

Secure Chat System 2025

Python, MySQL, AES-128, RSA, Diffie-Hellman, X.509. Code: github.com/AhmedMufti/Secure-Chat-System

- Client-server chat giving confidentiality, integrity, authenticity, and non-repudiation: certificate-based authentication against a local CA, Diffie-Hellman session keys, AES-128 message encryption, and RSA signatures.

EXPERIENCE

Freelance Software Engineer, Fiverr (Level 1 Seller) Jun 2024 - Present

Independent clients, remote

- Delivered backend and full-stack projects for international clients, from data model and API design to deployment, and reached Level 1 Seller through consistent on-time delivery.

Cyber Security Intern Jul 2025 - Aug 2025

Risk Associates, Islamabad, Pakistan

- Ran penetration tests and security audits on enterprise web applications and reported OWASP Top 10 vulnerabilities with remediation guidance.

Teaching Assistant (Part-time) Jan 2025 - Jun 2026

FAST-NUCES, Islamabad, Pakistan

- Ran labs, wrote assignments, and graded projects for Computer Networks, Theory of Automata, and Professional Practices in IT, supporting over 100 students; selected on academic performance.

EDUCATION

BSc Computer Science Aug 2022 - Jun 2026

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan

- CGPA 3.6 / 4.00. Dean's List for four consecutive semesters (Spring 2024 to Fall 2025). Coursework: Database Systems, Operating Systems (A+), Computer Networks (A), Information Security (A+), Software Engineering (A+), Parallel and Distributed Computing, Design and Analysis of Algorithms.

ACHIEVEMENTS & CERTIFICATIONS

- Sole-author research study of small language models in agentic RAG (manuscript in Springer LNCS format; MIT code on GitHub). Dean's List, four consecutive semesters. Fiverr Level 1 Seller. IELTS Academic 8.0. Machine Learning Specialization (Stanford / Coursera, in progress); Pre Security (TryHackMe).